LASI WAVE 1

Complete Project Blueprint

Early-Life Disadvantage and Later-Life Cognitive / Mental Health

A 14-day, survey-weighted quantitative ageing study using LASI Wave 1 (2017-18)

Project objective

Build a reproducible, cross-sectional epidemiological analysis testing whether childhood socioeconomic and health disadvantage is associated with later-life cognitive performance and depressive symptoms, and whether associations differ across age groups.

Prepared as a student research execution plan

Version: 04 October 2026

Executive Summary

This document turns the selected LASI research idea into an end-to-end project that a student team can execute in approximately two weeks. It starts from the data request and continues through data auditing, variable construction, survey-weighted estimation, interpretation, figures, writing, and final reproducibility checks.

The proposed design is a cross-sectional secondary analysis of LASI Wave 1 (2017-18). LASI Wave 1 is a nationally representative ageing survey covering adults aged 45 years and above. The uploaded LASI literature confirms that childhood socioeconomic status, childhood health, childhood bedridden experience, depression, cognition, demographic characteristics, socioeconomic status and health variables are available in Wave 1.

Non-negotiable interpretation rule

Because Wave 1 is a cross-sectional dataset for this project, the study estimates associations and age-group differences. It does NOT estimate within-person cognitive decline, incidence, or causal effects over time.

The primary scientific idea is a life-course question: whether disadvantages reported for childhood are associated with mental and cognitive outcomes measured decades later. The differentiating analytical feature is age heterogeneity - testing whether the association is different among 45-59, 60-74 and 75+ respondents.

Recommended primary outcome: continuous LASI cognitive score. Secondary cognitive outcome: binary poor cognition using the LASI literature-defined lower tail threshold after verifying the exact coding in the codebook. Secondary mental-health outcome: depressive symptoms using the ten-item CES-D threshold of 4 or more.

Recommended primary model: survey-weighted linear regression for the continuous cognitive score and survey-weighted modified Poisson regression with robust variance for binary outcomes, with logistic regression as a sensitivity analysis. All models must account for sampling weights, strata and PSU/cluster identifiers.

Project Map

Stage	Main question	Output
0. Define	What exactly are we testing?	Frozen protocol + hypotheses
1. Request	What files are needed?	Wave 1 .dta + codebook/questionnaires
2. Audit	What is inside the dataset?	Data dictionary + variable inventory
3. Clean	Who is eligible and how are variables coded?	Analysis-ready dataset
4. Describe	What does the sample look like?	Table 1 + prevalence tables
5. Model	Are exposures associated with outcomes?	Regression tables
6. Interaction	Does association differ by age?	Interaction tests + predicted probabilities
7. Robustness	Would the conclusion survive reasonable alternatives?	Sensitivity analyses

Stage	Main question	Output
8. Interpret	What can we claim?	Results + limitations
9. Report	Can another student reproduce it?	Report + code + data dictionary

1. Final Research Question and Scope

1.1 Primary research question

Among Indian adults aged 45 years and above, are adverse childhood socioeconomic and health conditions associated with lower cognitive performance in later life, and does the magnitude of this association differ across age groups?

1.2 Secondary question

Are the same childhood exposures associated with depressive symptoms, and do the age patterns look similar for mental and cognitive outcomes?

1.3 Core hypotheses

- Poorer perceived childhood socioeconomic status is associated with poorer cognitive performance.
- Poorer reported childhood health is associated with poorer cognitive performance.
- Having been bedridden for at least one month before age 16 is associated with poorer cognitive performance.
- The association between childhood disadvantage and cognitive/mental outcomes differs across age groups (effect modification by age group).
- Associations may attenuate after adjustment for current adult socioeconomic conditions, because education and current economic resources may lie on the life-course pathway.

1.4 Scope boundaries

- No claim of causality from Wave 1 alone.
- No claim of longitudinal cognitive decline.
- No merging with the 2010 pilot for the primary analysis.
- No specialized biomarker, spirometry or environmental files for the core analysis.
- No machine-learning prediction as the primary method; this is an epidemiological association study.

2. Dataset: What to Obtain Before Any Analysis

2.1 Minimum data package

Item	Required?	Why
LASI Wave 1 State & UT respondent-level microdata (.dta)	YES	Core individual-level observations
LASI Wave 1 Codebook / data dictionary	YES	Exact variable names, labels, coding, missing-value conventions
LASI Wave 1 Individual Schedule / questionnaire	YES	Verifies wording and construction of survey measures

Item	Required?	Why
Sampling design documentation	YES	Needed to identify weight, strata and PSU variables
Household file + link key	Preferable / verify	Needed if required economic variables are not already merged into the analytic file
Metro Cities data	NO	Not needed for an all-India project
DBS / Spirometry / Environmental Exposure	NO	Outside the core research question
LASI Pilot 2010	NO	Not needed for the primary cross-sectional Wave 1 analysis

Data request wording

Request the State & UT Wave 1 microdata in Stata (.dta) format with the corresponding codebook/questionnaire and survey-design identifiers. State explicitly that the project requires respondent-level variables for childhood SES, childhood health, childhood bedridden history, cognition, depression, age, demographics, current SES, health and survey weights/PSU/strata.

2.2 Why .dta?

The .dta format is Stata format. The .sav format is SPSS format. They are alternative statistical-software representations of data, not two distinct surveys. For this project, use the .dta version unless the data provider gives a reason to use another format.

3. Variable Inventory: What We Need and Why

3.1 Primary exposures

Variable	Operational role	Expected coding / construction	Priority
Childhood socioeconomic status	Primary exposure	Poor / Average / Pretty well-off, based on childhood family financial situation	MUST
Childhood health	Primary exposure	Original 5-level response; main analysis may use published binary grouping after verification	MUST
Childhood bedridden >=1 month	Secondary exposure / robustness	Yes / No	MUST
Parental education	Alternative early-life SES proxy	Mother/father education categories if available	HIGH

3.2 Outcomes

Outcome	Type	How to use	Priority
Cognitive score	Continuous	Primary outcome; retain original score and inspect distribution	MUST
Cognitive impairment / poor cognition	Binary	Secondary outcome; use published lower-tail threshold after codebook verification	HIGH
Depressive symptoms (CES-D)	Binary	Secondary outcome; 10-item score >=4 if complete score is reconstructed and verified	MUST
CES-D score	Discrete/continuous-like	Sensitivity / descriptive outcome	HIGH

3.3 Demographic and socioeconomic covariates

- Age in years - retain continuous age; construct 45-59, 60-74, 75+ for figures and interaction interpretation.
- Sex/gender - use the LASI survey variable and document how it is labelled.
- Residence - rural / urban.
- Caste / social group - use LASI categories; avoid inventing a new aggregation until frequencies are inspected.
- Marital status - preferably retain detailed categories first, then collapse only if analytically justified.
- Education - years of schooling if available; otherwise validated attainment categories.
- Current household economic status - preferably MPCE/consumption quintile or the LASI-provided economic measure.
- State/UT and region - for geographic adjustment / descriptive heterogeneity; avoid automatically using state fixed effects without checking the model design.

3.4 Health and social covariates

- Chronic conditions / multimorbidity - useful as current health context and sensitivity adjustment.
- Self-rated health - useful in secondary models; be cautious about over-adjustment because it may reflect pathways from early-life disadvantage to current health.
- ADL and IADL - functional health measures; secondary controls only unless explicitly justified.
- Physical activity, tobacco and alcohol - optional behavioural confounders / sensitivity controls.
- Living arrangement, social participation and other family/social variables - optional secondary context, not core exposures.

4. Technical Terms: Student-Level Definitions

Microdata: The dataset where each row is an individual/household and columns are measurements. This is different from a factsheet, where numbers are already aggregated.

Cross-sectional study: A study where exposure and outcome are observed in the same survey round. It can estimate associations but does not directly observe change within the same person over time.

Outcome / dependent variable: The phenomenon we are trying to explain or measure, such as cognitive score or depressive symptoms.

Exposure / independent variable: The characteristic whose association with the outcome is being estimated, such as childhood SES.

Covariate: A variable included in a statistical model for adjustment, description or precision.

Confounder: A variable associated with both the exposure and outcome that can create a misleading association if omitted. Adjustment is intended to reduce confounding.

Mediator: A variable on the pathway between exposure and outcome. Adjusting for a mediator can remove part of the association being studied. Adult health may be a mediator in a life-course hypothesis.

Effect modification / interaction: The association between exposure and outcome changes depending on another variable. Here, the main question is whether the childhood-disadvantage association differs by age group.

Sampling weight: A number used to compensate for unequal probabilities of selection and survey nonresponse so estimates better represent the target population.

PSU / cluster: Primary Sampling Unit or cluster used in the survey design. Responses within the same cluster may be correlated.

Stratum: A sampling layer used by the survey design. Survey software uses strata with weights and PSU to calculate design-correct standard errors.

Standard error (SE): A measure of statistical uncertainty around an estimated coefficient. Smaller SE generally means a more precise estimate.

95% confidence interval (CI): A range expressing sampling uncertainty around an estimate. For ratios, a CI crossing 1 often indicates weak evidence for an association at the conventional 5% level.

P-value: A measure used in frequentist testing. It should not be interpreted as the probability that the hypothesis is true.

Odds ratio (OR): Ratio of odds between groups. It is not the same as a risk ratio or prevalence ratio, especially when the outcome is common.

Prevalence ratio (PR): Ratio of outcome prevalence between exposure groups. It is often easier to interpret than an OR for common cross-sectional outcomes.

Modified Poisson regression: A Poisson regression model with robust variance often used to estimate prevalence ratios for binary outcomes in cross-sectional data.

Linear regression: Regression where a continuous outcome is modelled as a weighted sum of predictors plus error; here it can model cognitive score.

Robust standard error: A standard-error estimator designed to remain valid under certain forms of variance misspecification. In survey analysis, the full survey design should still be specified.

Multicollinearity: A situation where predictors are strongly correlated with each other, making individual coefficients unstable. VIF is one common diagnostic.

Complete-case analysis: Analyzing respondents with nonmissing values on all variables in a model. Easy, but it can change the study population and introduce bias if missingness is systematic.

Sensitivity analysis: A secondary analysis using a reasonable alternative definition/model to test whether the conclusion is robust.

Predicted probability: The model-based estimated probability of an outcome for a chosen set of covariates. Useful for interpreting interactions.

5. Conceptual Framework and Causal Language

The study is motivated by a life-course framework: conditions in childhood may be associated with accumulated educational, socioeconomic, behavioural and health pathways that are reflected in later-life cognition and mental health. The model is used as a conceptual guide; it does not turn a cross-sectional association into a causal effect.

5.1 Primary conceptual pathway

Node	Meaning in this project
Childhood SES / childhood health	Early-life exposures reported retrospectively by respondents
Education	Potential long-term pathway and current socioeconomic marker; treat carefully in sequential models
Adult economic status	Current resources / material conditions
Adult health	Potential confounder for some social associations, but also potentially downstream of early life
Cognitive score / depression	Later-life outcomes observed in Wave 1
Age group	Potential effect modifier for cross-sectional heterogeneity

Core analytical principle

Build the primary model from variables plausibly preceding the outcome and pre-specify a second model that adds adult socioeconomic variables. Do not call the difference between models "mediation" unless a formal mediation analysis is conducted.

6. Data Acquisition, Import and Audit

6.1 Folder structure

- 01_raw/ - untouched LASI files received from IIPS
- 02_reference/ - codebook, questionnaires, data documentation, paper PDFs
- 03_clean/ - derived analysis-ready data
- 04_code/ - Stata/R/Python scripts
- 05_output/ - tables, figures, model objects
- 06_report/ - manuscript and presentation

Rule

Never edit the raw .dta file directly. Read it, document it, and create a derived clean dataset. This protects reproducibility.

6.2 First-hour audit checklist

1. Open the data and report number of rows and columns.
2. Inspect variable names, labels and value labels.

3. Check that age, respondent ID and household ID exist.
4. Locate survey weights, PSU and strata variables.
5. Locate every childhood, cognition and depression variable before creating derived variables.
6. Inspect missing-value conventions. LASI may use special codes for don't know, not applicable or missing; these must not be treated as real numerical values.
7. Check duplicate respondent IDs.
8. Check whether household and individual data are separate files and determine the correct merge key before merging.
9. Create a one-page variable inventory with: variable name, label, source question, coding, missing values, analysis role.

7. Sample Definition and Data Cleaning

7.1 Eligibility

Start from all LASI Wave 1 respondents eligible for the individual ageing analysis. Apply the pre-specified age criterion of 45 years and above. Do not delete records merely because a variable is missing until you know whether that variable is required for a particular outcome/model.

7.2 Create a participant flow

Stage	N	What to record
Raw Wave 1 individual records	TBD after import	Raw row count
Age >=45	TBD	Primary target population
Valid childhood SES	TBD	Missingness
Valid childhood health	TBD	Missingness
Valid cognitive assessment	TBD	Outcome-specific sample
Valid CES-D / depression	TBD	Outcome-specific sample
Complete primary model covariates	TBD	Model-specific complete-case sample

7.3 Missing values

Do not automatically recode every nonpositive number as missing. First inspect the codebook. Create explicit missing indicators only for diagnosis/documentation; do not use missingness as a substantive exposure unless the protocol says so.

7.4 Outlier and range checks

- Age: verify plausible range and inspect extreme values.
- Cognitive score: verify the theoretical minimum/maximum against the LASI scoring system.
- CES-D: verify each item coding and reverse-coded positive-affect items before summing.
- MPCE: inspect skewness; if using quintiles supplied by LASI, document the source rather than reconstructing unnecessarily.

8. Variable Construction: Exact Workflow

8.1 Childhood SES

10. Locate the exact childhood financial-status question in the questionnaire and codebook.
11. Verify that response labels correspond to Poor, Average and Pretty well-off.
12. Preserve the raw categorical variable as the audit variable.
13. Create a labelled analysis variable with Pretty well-off as the reference category.
14. Check weighted prevalence in each category. If any category is extremely small, discuss before collapsing.

8.2 Childhood health

15. Locate the five-level childhood health question.
16. Preserve the original 5-level measure.
17. For the main model, use the published binary grouping only after checking that the LASI version and coding match the published paper.
18. Run a sensitivity analysis using the original ordered categories if cell sizes permit.

8.3 Childhood bedridden experience

19. Code No=0 and Yes=1 using the LASI labels.
20. Confirm that the question refers to before age 16 and at least one month.
21. Report weighted prevalence because this exposure is expected to be uncommon.

8.4 Cognitive score

22. Identify all cognitive test components and/or the LASI composite score.
23. Verify scoring direction: higher score should represent better performance.
24. Inspect distribution by age and education before selecting transformations.
25. Use continuous score as the preferred primary outcome if its construction is valid.
26. Construct poor cognition using the published lower-tail definition after verifying the exact LASI implementation.

8.5 CES-D

27. Identify the ten items and response-frequency coding.
28. Recode negative symptom items to 0/1 according to the LASI/published scoring rule.
29. Reverse-score positive-affect items as instructed by the scale.
30. Sum the ten items to obtain a 0-10 score if missingness rules are satisfied.
31. Create depressive symptoms = 1 for score >=4, 0 otherwise, if using the published Wave 1 operationalization.

9. Quantitative Analysis Plan

9.1 Analysis order

Analysis	Purpose	Main output
A. Weighted descriptives	Describe the target population	Table 1
B. Weighted bivariate	Show unadjusted patterns	Prevalence/means by exposure
C. Model 1	Demographic adjustment	Adjusted association

Analysis	Purpose	Main output
D. Model 2	Add adult SES	Assess attenuation / persistence
E. Model 3	Add selected health variables	Context / robustness
F. Interaction model	Test age heterogeneity	Interaction p-value + predicted values
G. Sensitivity	Check alternative specifications	Robustness table

9.2 Descriptive statistics

- Use survey weights for population percentages, means and confidence intervals.
- Describe age, sex, residence, education, economic status and each primary exposure.
- Report mean cognitive score and prevalence of poor cognition/depressive symptoms overall and by age group.
- Do not infer causality from descriptive differences.

9.3 Primary continuous cognitive model

Recommended model: survey-weighted linear regression with cognitive score as the outcome. Conceptually:

CognitiveScore = beta0 + beta1(ChildhoodSES) + beta2(ChildhoodHealth) + beta3(Age) + beta4(Sex) + ... + error

Interpretation: beta for an exposure category is the adjusted difference in mean cognitive score relative to the reference category, conditional on included covariates and the survey design.

9.4 Binary outcome models

For poor cognition and depressive symptoms, use survey-weighted modified Poisson regression with robust variance when the outcome is common enough that odds ratios would be less intuitive. Report adjusted prevalence ratios (APR) and 95% CIs. Run logistic regression as a sensitivity analysis and report AORs separately.

log(E[Y]) = beta0 + beta1X1 + ... + betakXk

For a binary outcome under a modified Poisson model, exp(beta) is interpreted as an adjusted prevalence ratio under the chosen specification.

9.5 Model blocks

Model	Variables	Scientific purpose
M0: crude	One exposure only + survey design	Unadjusted association
M1: demographic	Exposure + age + sex + residence + caste + marital status	Control major demographic confounding
M2: socioeconomic	M1 + education + current economic status	Ask whether early-life association persists after adult SES adjustment
M3: health context	M2 + selected chronic/functional health variables	Robustness / current health context; interpret cautiously because some variables may be downstream

10. Age Interaction: The Most Important Extension

Age interaction asks whether the association between childhood disadvantage and the outcome is different for people at different ages. This is not the same as showing that a person deteriorates with age.

10.1 Specification

Create age groups 45-59, 60-74 and 75+. Fit a model containing the exposure, age-group indicators, and exposure-by-age-group interaction terms.

Outcome = beta0 + beta1(ChildhoodSES) + beta2(AgeGroup) + beta3(ChildhoodSES x AgeGroup) + Covariates

10.2 How to test it

32. Fit the main-effects model.
33. Add the interaction terms.
34. Use a joint test of the interaction terms rather than interpreting one interaction coefficient in isolation.
35. Generate predicted probabilities or predicted mean cognitive scores for each age-by-exposure combination.
36. Plot the estimates with 95% confidence intervals.

Example interpretation

Suppose the adjusted prevalence ratio for poor cognition associated with poor childhood SES is higher in the 75+ group than in 45-59. The appropriate statement is: "The association was stronger among adults aged 75 years and above." Do not write "poor childhood SES caused faster cognitive decline."

11. Confounding, Mediation and Over-Adjustment

11.1 Why this matters

Students often put every available variable into one regression. That is not automatically better. Some variables may be consequences of the exposure and therefore lie on the pathway to the outcome.

11.2 Practical decision rule

Variable	Primary role	Why
Age	Core covariate + effect modifier	Strongly related to cognition and outcome risk
Sex	Core covariate	Demographic and biological/social differences
Education	Sequential adjustment variable	May partly reflect life-course pathway from childhood conditions
Current MPCE/economic status	Sequential adjustment variable	Captures current material resources
Caste/residence/marital status	Core demographic covariates	Social and demographic context
Multimorbidity	Secondary health adjustment	May be confounder for some associations but also downstream
ADL/IADL	Secondary/sensitivity	Could partly reflect pathways to cognitive/mental health
Self-rated health	Secondary/sensitivity	Strong health marker but potentially

Variable	Primary role	Why
		downstream
Depression in cognition model	Usually secondary only	Could be correlated with cognition and partly share mechanisms; do not automatically adjust in the primary model

12. Survey Design: How to Get Nationally Correct Inference

LASI uses a multistage, stratified probability sampling design. Standard unweighted regression treats the sample as if it were a simple random sample, which is not the design LASI used. The analysis should use the provided survey weights and design identifiers.

12.1 Three core design elements

Element	Plain-language meaning	What it changes
Weight	Who each respondent represents	Population estimates / point estimates
PSU	Which sampled cluster the person came from	Variance estimation
Strata	Sampling layer	Variance estimation / design representation

12.2 Software

Stata is a natural choice because the requested data are .dta and Stata has mature survey commands. R can also perform the analysis using the survey package or equivalent tooling. The team should choose one primary software path and avoid duplicating every analysis in two languages during the two-week window.

Software pattern

Stata concept: define the survey design once, then use survey-aware summary and regression commands. R concept: create a survey design object, then use svymean / svyglm-style procedures. Exact variable names and commands depend on the received codebook.

13. Model Diagnostics and Quality Checks

13.1 Before trusting a regression

- Check whether the analysis sample changed unexpectedly between models.
- Check frequencies for every categorical predictor.
- Check the outcome prevalence and score distribution.
- Check missingness before model fitting.
- Check VIF / correlation among continuous or highly related predictors where appropriate.
- Inspect whether estimates are unstable because a category is too small.
- Check convergence and warnings from the statistical software.

- Compare a weighted and unweighted model only as a diagnostic; do not replace the design-correct model with the unweighted one.

13.2 Cognitive-score checks

- Plot cognitive score distribution.
- Compare mean score across age and education groups.
- Check whether a simple linear specification is reasonable; if heavily non-linear, consider age splines or categorized age in secondary analyses.

13.3 Binary outcome checks

- Verify that the outcome is coded only 0/1 in the final model.
- Ensure missing, not applicable and don't know are not interpreted as outcome=0.

14. Sensitivity Analyses

Sensitivity	Alternative	Why
S1	Continuous cognition vs poor-cognition indicator	Tests dependence on a cutoff
S2	Original 5-level childhood health vs binary grouping	Tests dependence on recoding
S3	Crude -> M1 -> M2 -> M3 sequential models	Shows how adjustment changes the estimate
S4	Logistic regression vs modified Poisson	Shows whether interpretation depends on effect measure
S5	Age continuous vs age groups	Tests whether age categorization drives the interaction
S6	Complete-case vs carefully documented alternative missing-data approach if feasible	Assesses missing-data sensitivity

Time management

Do not perform every sensitivity analysis automatically. S1-S4 are the minimum useful set. S5-S6 are optional depending on the data quality and remaining time.

15. Tables and Figures to Produce

15.1 Tables

Table	Content	Decision
Table 1	Weighted characteristics of respondents overall and by age group	Descriptive
Table 2	Weighted outcome means/prevalence by childhood exposure	Unadjusted

Table	Content	Decision
Table 3	Cognitive-score regression (M1-M3)	Primary result
Table 4	Poor cognition APR/AOR (M1-M3)	Secondary cognitive result
Table 5	Depressive-symptom APR/AOR (M1-M3)	Secondary mental-health result
Table 6	Age-interaction / joint tests	Key novelty
Table 7	Sensitivity analysis	Robustness

15.2 Figures

- Figure 1: Project conceptual framework / DAG.
- Figure 2: Mean cognitive score across age groups by childhood SES.
- Figure 3: Adjusted predicted probability of poor cognition by age group and childhood SES, with 95% CI.
- Figure 4: Adjusted predicted probability of depressive symptoms by age group and childhood SES, with 95% CI.

Figure principle

Prefer predicted values from the fully specified survey-weighted model for interaction results. Raw prevalence is useful for descriptive figures; adjusted predicted values are better for communicating the modelled age interaction.

16. How to Interpret the Numbers

16.1 Linear regression example

Suppose beta = -1.20 for poor childhood SES. The interpretation would be: respondents in the poor childhood-SES category had, on average, 1.20 points lower cognitive score than the reference group, conditional on the variables in the model. This is an association, not a causal effect.

16.2 Prevalence ratio example

Suppose APR = 1.25 for poor childhood health. The interpretation would be: the prevalence of the binary outcome was estimated to be about 25% higher in the poor-childhood-health group than in the reference group, conditional on the model covariates and survey design.

16.3 Confidence interval example

Suppose APR = 1.25 with 95% CI 1.10-1.42. The estimate suggests a positive association and the interval excludes 1.00. Report the estimate and uncertainty, not only the p-value.

16.4 Non-significant interaction

If the interaction test is not statistically persuasive, do not manufacture separate "effects by age" conclusions. Report that strong evidence of heterogeneity by age was not detected, while still presenting descriptive patterns if informative.

17. Final Research Report Structure

Section	What it must contain
Title	Early-Life Socioeconomic and Health Disadvantage and Cognitive and Mental Health in Ageing India: Evidence from LASI Wave 1, 2017-18
Abstract	Background; Methods; Results; Conclusion. Keep causal language out.
Introduction	Ageing in India; why cognition/mental health matter; life-course rationale; prior LASI evidence; gap; objective.
Methods	Data source; sample; variables; outcome construction; exposure construction; survey design; statistical models; sensitivity analyses; ethics/data access statement.
Results	Sample flow; descriptive table; outcome prevalence; primary regression; interaction; sensitivity.
Discussion	Main findings; comparison with literature; plausible mechanisms; interpretation of age heterogeneity; policy implications; strengths/limitations.
Conclusion	One paragraph, no causal overreach.
References	LASI documentation and peer-reviewed sources.
Supplementary material	Variable map, missingness table, code list, additional model diagnostics.

18. Fourteen-Day Execution Plan - Minute-Level Version

The schedule below assumes the Wave 1 data arrive by Day 1. If access is delayed, shift the literature-writing tasks earlier and preserve the analysis days.

Day	Morning (approx. 3h)	Afternoon (approx. 3-4h)	Deliverable
1	Freeze research question, outcomes, exposures; read codebook sections	Import raw .dta; inspect rows/columns, IDs, labels, missing codes, survey design	Protocol v1 + raw-data audit
2	Build variable inventory and map questionnaire questions	Locate childhood, cognition, depression variables; resolve file merges	Variable dictionary
3	Code age, demographics, childhood exposures	Construct cognition and CES-D measures; verify score ranges	Clean derivation script
4	Check missingness, duplicates, range errors	Finalize analytic samples; generate participant flow counts	Analysis-ready data v1
5	Set survey design; test weighted descriptives	Table 1 + weighted outcome prevalence	Descriptive results
6	Run crude models	Run M1 demographic models	Regression draft v1
7	Run M2 current-SES models	Run M3 health-context models; diagnostics	Primary model set

Day	Morning (approx. 3h)	Afternoon (approx. 3-4h)	Deliverable
8	Run age interactions	Joint tests + predicted values	Interaction results
9	Sensitivity S1-S4	Check robustness and model stability	Sensitivity table
10	Produce figures	Clean table/figure formatting; freeze analysis output	Tables + figures v1
11	Write Methods	Write Results directly from frozen outputs	Methods + Results draft
12	Write Introduction	Write Discussion linked to evidence and limitations	Full manuscript draft
13	Audit references, tables and claims	Reproducibility run from raw/clean data to outputs	QA checklist complete
14	Final editing	Presentation, executive summary, code packaging	Final report + code archive

18.1 Daily operating rhythm

- First 15 min: write the exact question for the day.
- Next 2-3 h: execute one analysis task only; save code and output.
- 30 min: inspect results and document anomalies.
- 30-60 min: clean tables/notes, not the raw data.
- Final 15 min: update the project log with what changed and what remains.

19. Reproducibility and Team Workflow

19.1 Freeze the analysis

Once the main descriptive checks and variable definitions are confirmed, write the final analysis specification in a one-page protocol. After that point, changing hypotheses because a result is significant is not allowed.

19.2 File naming

- raw_lasi_wave1.dta
- variable_inventory.xlsx or .csv
- clean_lasi_topic1.dta
- 01_import.do / 01_import.R
- 02_construct.do / 02_construct.R
- 03_descriptive.do / 03_descriptive.R
- 04_models.do / 04_models.R
- 05_figures.do / 05_figures.R
- tables/ figure/ outputs folders

19.3 Minimum analysis log

Date	Change	Reason	Impact
YYYY-MM-DD	Example: changed childhood-health coding	Codebook verification	Affects models M1-M3
YYYY-MM-DD	Example: excluded invalid	Data validation	Changes cognition sample N

Date	Change	Reason	Impact
	cognitive score		
YYYY-MM-DD	Example: retained age continuous in primary model	Protocol choice	Interaction uses age groups

20. Ethics, Data Governance and Reporting

The analysis is secondary research using de-identified survey data. Follow the data provider's terms of use and the research ethics requirements of the institution. Do not attempt to identify individuals or households. In the report, describe the original LASI ethics/consent framework and the permissions applicable to secondary use, as documented by the data provider.

- Do not publish raw respondent identifiers.
- Do not upload restricted microdata to public repositories.
- Store the raw file separately from public code.
- Share derived aggregate tables/figures only according to LASI data-use terms.

21. Common Failure Modes and the Prevention Plan

Failure	Why it is bad	Prevention
Treating a factsheet as the dataset	Aggregate numbers cannot support individual regression	Use respondent-level .dta microdata
Ignoring survey weights	Population estimates and standard errors can be wrong	Define survey design before inferential analysis
Using ordinary logistic regression only	OR can be hard to interpret for common outcomes; design ignored if no survey setup	Use survey-aware methods; present PR as primary for binary outcomes
Calling Wave 1 longitudinal	One wave does not observe within-person change	Use cross-sectional wording
Putting every variable in the model	Can over-adjust for mediators and create unstable estimates	Use a pre-specified sequential model strategy
Changing the cutoff after seeing results	Introduces outcome-driven analysis	Freeze operational definitions from codebook/literature
Ignoring missing codes	Don't know can become a fake score of zero	Inspect codebook and recode explicitly
Reporting only p-values	Hides magnitude and uncertainty	Report estimate + 95% CI + p-value where useful
Over-interpreting interactions	Random subgroup noise can look meaningful	Use joint tests + predicted values + CIs
Trying too many subgroups	Two-week project becomes unmanageable	Limit to pre-specified age and optional sex stratification

22. Final Deliverables Checklist

- One-page protocol with final research question and hypotheses.

- Clean variable dictionary mapping every analysis variable to the LASI source question and exact data variable name.
- Clean analysis dataset derived from raw data.
- Complete reproducible code from import to tables and figures.
- Weighted Table 1 and outcome prevalence tables.
- Primary cognitive regression table.
- Binary outcome regression tables for poor cognition and depression.
- Age-interaction table and predicted-value figures.
- Sensitivity analysis table.
- Final research report with limitations and no causal overclaiming.
- Five-minute oral explanation of the design, variables, model and key finding.

23. What If Nothing Is Significant?

A null result is not a failed project. A rigorous study can find weak or absent associations. The correct response is to report the magnitude, confidence interval and plausible explanations: measurement error, retrospective reporting of childhood conditions, residual confounding, heterogeneity across cohorts, selective survival, or insufficient variation in the exposure. Do not "search" for a significant subgroup until one appears.

24. Minimum Viable Project (MVP) if Time Runs Out

The 2-day emergency version

If the team loses time, keep only: (1) one childhood exposure - childhood SES; (2) primary outcome - continuous cognitive score; (3) age groups 45-59 / 60-74 / 75+; (4) age interaction; (5) core demographic + current SES adjustment; (6) survey-weighted linear regression; (7) one predicted-value figure. Add depression and sensitivity analyses only after this core analysis is complete.

25. Source Basis and Literature Anchors

The project design is informed by the uploaded LASI papers, especially the 2024 study that operationalized childhood SES, childhood health, childhood bedridden experience and CES-D depressive symptoms, and the 2026 study that operationalized LASI cognition and used age groups 45-59, 60-74 and 75+ alongside survey-weighted regression. These papers are used to define feasible LASI measures, not to claim that the proposed project is longitudinal or causal.

[1] Khanal G, Selvamani Y. Association of perceived childhood socio-economic status and health with depressive symptoms among middle-aged and older adults in India: using data from LASI Wave I, 2017-2018. BMC Geriatrics. 2024;24:226. doi:10.1186/s12877-024-04800-0.

[2] Sharma M, Goswami I. Is all movement equal at every age? Differentiating the impact of physical activity types on depressive symptoms and mental acuity in an aging India. Archives of Public Health. 2026;84:133. doi:10.1186/s13690-026-01943-x.

[3] Rana GS et al. Association of multi-morbidity, social participation, functional and mental health with the self-rated health of middle-aged and older adults in India: a study based on LASI wave-1. BMC Geriatrics. 2022;22:675. doi:10.1186/s12877-022-03349-0.

[4] Srivastava S, Muhammad T. Socioeconomic vulnerability and frailty among community-dwelling older adults: cross-sectional findings from LASI, 2017-18. BMC Geriatrics. 2022;22:201. doi:10.1186/s12877-022-02891-1.

[5] Muhammad T et al. Socioeconomic and health-related inequalities in major depressive symptoms among older adults: a Wagstaff decomposition analysis of the LASI baseline survey, 2017-2018. BMJ Open. 2022;12:e054730. doi:10.1136/bmjopen-2021-054730.

[6] Lal D et al. Healthcare access and utilisation trends among the elderly in India: Evidence from the LASI Wave-1 survey. Dialogues in Health. 2025;7:100232. doi:10.1016/j.dialog.2025.100232.

[7] Mohanty SK et al. Sociodemographic and geographic inequalities in diagnosis and treatment of older adults chronic conditions in India: a nationally representative population-based study. BMC Health Services Research. 2023;23:332. doi:10.1186/s12913-023-09318-6.

[8] Srivastava S, Chauhan S, Patel R. Socio-Economic Inequalities in the Prevalence of Poor Self-Rated Health among Older Adults in India from 2004 to 2014: A Decomposition Analysis. Ageing International. 2021;46:182-199. doi:10.1007/s12126-020-09385-8.

[9] International Institute for Population Sciences (IIPS). LASI Wave I documentation and data access materials. Verify exact current download/version and codebook accompanying the data request.

26. Final One-Page Team Brief

The project in one line

Using nationally representative LASI Wave 1 data, estimate whether reported childhood socioeconomic and health disadvantage is associated with poorer later-life cognition and depression, and test whether those associations vary by age group.

Question	Answer
Population	LASI Wave 1 respondents aged 45+
Design	Cross-sectional secondary analysis
Primary exposure	Childhood socioeconomic status
Secondary exposures	Childhood health; childhood bedridden >=1 month
Primary outcome	Continuous cognitive score
Secondary outcomes	Poor cognition; depressive symptoms
Core age variable	45-59 / 60-74 / 75+ plus continuous age
Primary model	Survey-weighted linear regression
Binary outcome model	Survey-weighted modified Poisson + robust variance; logistic sensitivity
Key novelty	Exposure-by-age interaction / age heterogeneity
Must-have design variables	Weight + PSU + strata
Main danger	Cross-sectional design being misrepresented as longitudinal/causal
Target completion	14 days

Important: exact LASI variable names, missing-value codes, cognitive scoring details, sample weights, PSU and strata variables must be verified against the actual Wave 1 codebook when the data are received. The plan intentionally distinguishes what is already supported by the uploaded literature from items that must be confirmed in the raw data.